

# TIMOTHY YE

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## OBJECTIVE

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Mechanical Engineering student seeking a summer internship to apply mechanical and electrical engineering skills to hands-on hardware work. Fox Chapel Area High School alum and TSA graduate.

## EDUCATION

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### University of Pittsburgh — Swanson School of Engineering

Pittsburgh, PA

B.S. Mechanical Engineering | GPA: 3.13 / 4.00 | Minor in Electrical Engineering

Expected Graduation: May 2029

**Relevant Coursework:** Engineering Computation & Programming (MATLAB, C), Computer-Aided Design (SolidWorks), Engineering Analysis (Excel), First-Year Engineering: Sustainable Design

## TECHNICAL SKILLS

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**Design / CAD:** SolidWorks, 3D printing/additive design

**Programming & Analysis:** MATLAB, C, Microsoft Excel (data analysis)

**Fabrication:** Laser cutting/engraving, wood & metal shop tools, hand/power tools, spray finishing

**Electrical:** Basic circuit assembly & wiring, introductory avionics exposure

## ENGINEERING PROJECTS

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### Triboelectric Nanogenerator (TENG) Energy-Harvesting Tiles

Fall 2025

First-Year Engineering Course — Energy & Sustainability Project

- Researched, sketched, and built a working proof-of-concept TENG floor tile that converts mechanical footstep pressure into electrical energy
- Wired a basic circuit to power an LED on contact and presented the design and prototype to the class

### Piezoelectric Actuators for Morphing-Wing Aircraft

Spring 2026

First-Year Engineering Research Paper & Presentation

- Co-authored a 6,000-word research paper evaluating piezoelectric actuators as a lightweight, energy-efficient method of achieving morphing-wing control on aircraft
- Delivered a 10-minute technical presentation on findings and design implications to faculty and peers
- Built a C hangman game using a custom word bank drawn from the paper's technical terminology for a team mini-hackathon

## EXPERIENCE & ACTIVITIES

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### SOAR — Aerostructures Subteam, Member

Fall 2026 – Present

- Building CAD, manufacturing, and design skills while contributing to structural design and basic avionics/electrical integration for the team's aircraft builds

### Technology Student Association (TSA)

2023 – 2025

3-Year Member | State Qualifier all 3 years

- Year 1 — UAV Drone: Assembled and modified a UAV drone with a 3D-printed custom claw attachment; documented the build process for competition judging
- Year 2 — Board Game Design: Designed a laser-engraved board game; that same year, separately created a hand-bound, illustrated adventure storybook for the Children's Story event
- Year 3 — Manufacturing Prototype: Built a rotating picture frame / perpetual calendar using wood-shop manufacturing, laser engraving, and spray-paint finishing
- Year 3 — Flight Endurance: Built and flight-tested a kit-based aircraft, recording flight time, weight, and assembly data to optimize performance

### Lifeguard — American Pool (CSC & Fox Chapel Yacht Club)

2022 – 2025

- CPR/First Aid certified; 4 seasons
- Team Captain: Varsity Ultimate Frisbee (Jr., Sr. year) and Varsity Fencing (Sr. year)